Thrombus radiomics in patients with anterior circulation acute ischemic stroke undergoing endovascular treatment

Supplemental Materials

Table of contents

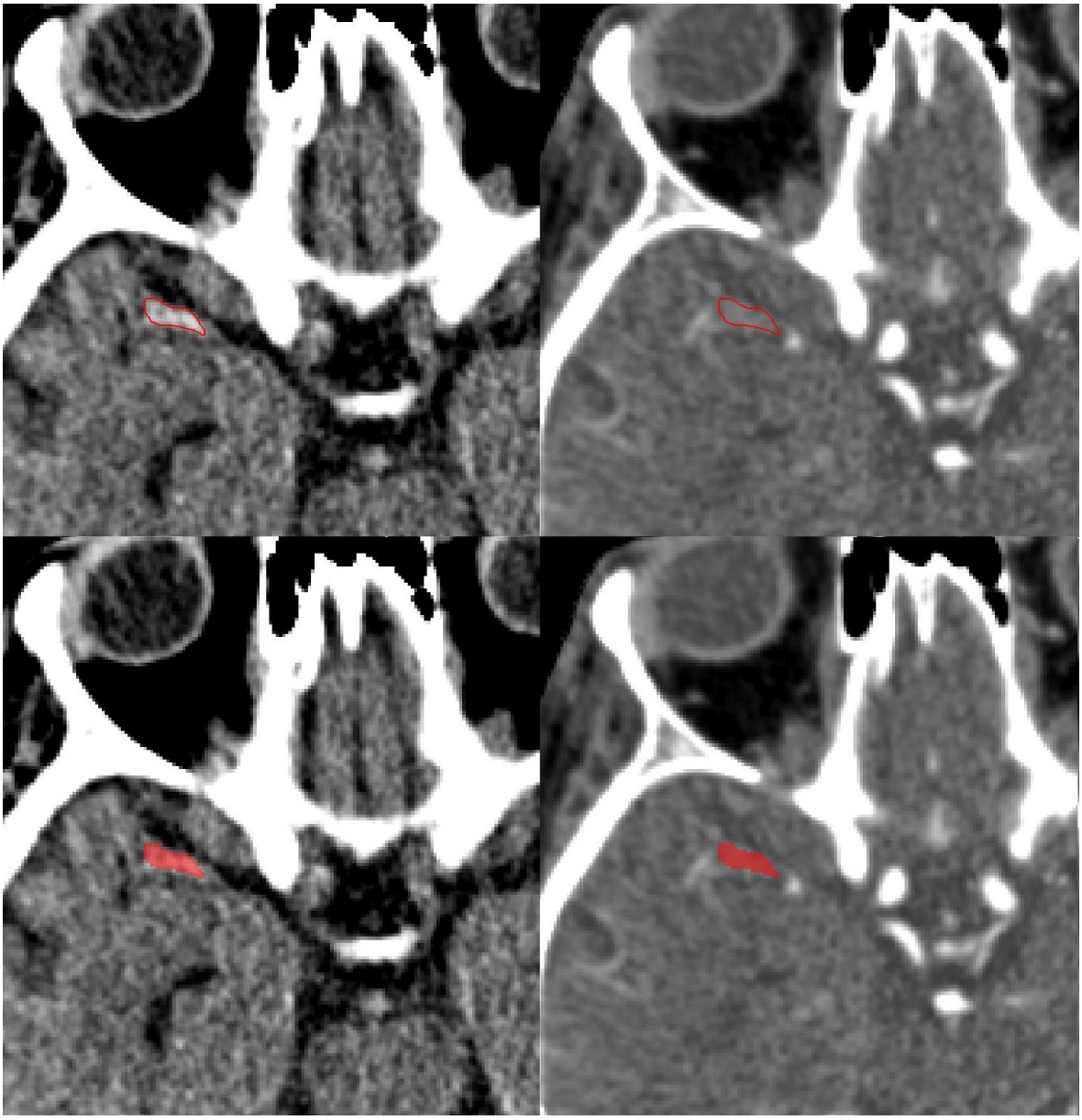
Supplemental Methods S1 3
Supplemental Figure S1 4
Supplemental Figure S2 6
Supplemental Table S1 7
Supplemental Table S2 8
Supplemental Table S3 8
Supplemental Table S4 11
Supplemental Table S5 14
Supplemental Table S6 15
Supplemental References 16

van Voorst H, *et al. J NeuroIntervent Surg* 2023; 15:e78–e78. doi: 10.1136/jnis-2022-019085

Supplemental Methods S1. Imaging analyses by the MR CLEAN Registry core laboratory

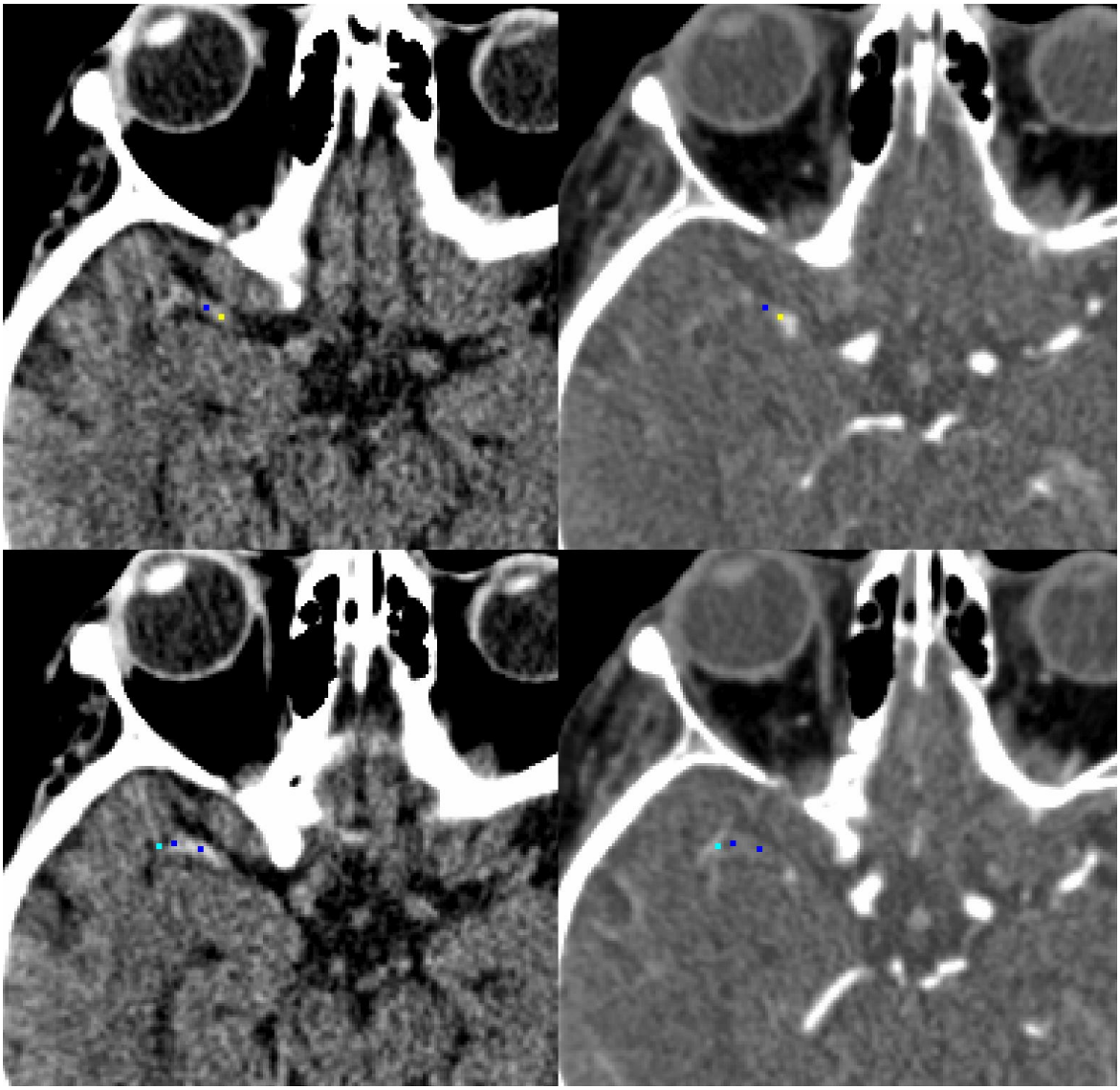
The following imaging characteristics were assessed by the MR CLEAN Registry imaging core lab, whose members were blinded to all clinical data except for symptom side:[1] Alberta stroke program early CT score (ASPECTS) on baseline NCCT,[2] collateral score on baseline CTA,[3] and reperfusion status on post-procedural digital subtraction angiography (DSA). Reperfusion status was evaluated according to the extended Thrombolysis in Cerebral Ischemia (eTICI) score.[4] Successful reperfusion was defined as eTICI ≥2B. Complete postintervention DSA imaging (anteroposterior and lateral views) was needed to be classified as successful reperfusion. If a lateral view was missing, an eTICI score of 2A was the highest possible score.

Supplemental Figure S1. Thrombus measurements and segmentation in the same patient. Non-contrast CT (left image, window width 80 HU and window level 40 HU) was co-registered with a CT angiography scan (right image window width 350 HU and window level 75 HU).



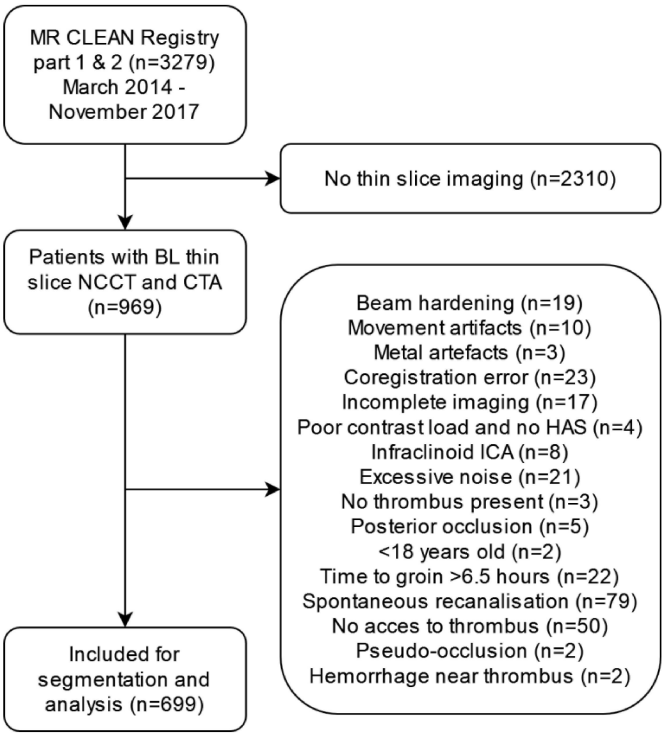
A. Full thrombus segmentation.

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B. Thrombus measurements. The yellow region of interest (ROI) indicates the proximal thrombus border, blue ROIs are placed inside the thrombus and the turquoise marker indicates the distal thrombus border.

Supplemental Figure S2. Patient inclusion flow-chart. Baseline (BL), Non-contrast CT (NCCT), CT with angiography (CTA). Hyperdense artery sign (HAS). Internal carotid artery (ICA)



Supplemental Table S1. Variable description

Variable	Variable type	Description
Age	Baseline clinical variable	Age in years
Sex	Baseline clinical variable	Male or female
Pre-stroke mRS	Baseline clinical variable	modified Rankin Scale score prior to acute stroke
NIHSS BL	Baseline clinical variable	Baseline National Institutes of Health Stroke Scale
Previous stroke	Baseline clinical variable	History of stroke prior to current stroke
Previous DM	Baseline clinical variable	History of diabetes (DM)
Previous AF	Baseline clinical variable	History of atrial fibrillation (AF)
Occlusion location	Baseline clinical variable	Site of occlusion location: Intracranial internal carotid artery (ICA), Intracranial internal carotid artery terminus (ICA-T), M1 segment of middle cerebral artery, M2 segment of middle cerebral artery, M3 segment of middle cerebral artery, A1, or A2 segment of anterior cerebral artery (categorical variable)
Occlusion location ICA	Baseline clinical variable	Site of the occlusion: Intracranial internal carotid artery (ICA)
Occlusion location ICA-T	Baseline clinical variable	Site of the occlusion: Intracranial internal carotid artery terminus (ICA-T)
Occlusion location M1	Baseline clinical variable	Site of the occlusion: M1 segment of middle cerebral artery
Occlusion location M2	Baseline clinical variable	Site of the occlusion: M2 segment of middle cerebral artery
Occlusion location other	Baseline clinical variable	Site of the occlusion: M3 segment of middle cerebral artery, A1, or A2 segment of anterior cerebral artery
Collateral score	Baseline clinical variable	Collateral score: 0% collateral filling of the occluded territory (0), >0 and ≤50% filling (1), >50 and<100% filling (2), 100% filling of the occluded territory (3)
ASPECTS BL	Baseline clinical variable	Alberta stroke programme early CT score at baseline non-contrast CT
Time to groin puncture	Baseline clinical variable	Time from onset of neurological deficit to groin puncture as start of the endovascular procedure in minutes
Systolic blood pressure	Baseline clinical variable	Systolic blood pressure at baseline in mmHg
Months since start	Baseline clinical variable	Months since start of the MR CLEAN Registry (included to adjust for increased experience of interventional neuroradiologists over time)
IVT	Baseline clinical variable	If intravenous alteplase treatment (IVT) was administered

Thrombus length	Manual thrombus measurement	Length of the thrombus from start to end of the filling defect on CT angiography (or hyperdense artery sign) in mm
Thrombus density	Manual thrombus measurement	Mean density in Hounsfield Units of 3 regions of interest (ROIs) in the thrombus
Perviousness	Manual thrombus measurement	Perviousness was calculated by subtracting mean density of the three non-contrast CT thrombus markers from the mean density of the CT angiography thrombus markers (attenuation increase in HU after intravenous contrast)

Supplemental Table S2. Hyper parameters used to optimize Random Forest models

Hyper parameter	**Values considered in grid search**
Minimum impurity decrease	1e-4, 7e-3, 5e-3, 3e-3, 1e-3, 1e-2
Evaluation criterion	Entropy
Maximum number of features considered per tree	1%, 5%, 10%, 15%, 20%, 30%, square root of n-vars, log-2 of n-vars
Minimum samples per split	25
Minimum samples per leaf	25
Class weight	Balanced, None
Bootstrap	TRUE
Number of trees	1000

For the presented work scikit-learn's ExtraTreesClassifier was used for the hyper parameters considered in the grid search. N-vars: number of variables used to predict outcome.

Supplemental Table S3. Baseline and treatment characteristics of the MR CLEAN Registry patients included in our study compared to the remainder of the MR CLEAN Registry population

	Remainder of MR CLEAN Registry population (n=2480)	**Training set (n=499)**	**Test set (n=200)**	**P value**
Registry Part				0.64
1 (March 2014-June 2016)	1150 (46.4%)	241 (48.3%)	97 (48.5%)	
2 (June 2016- November 2017)	1331 (53.6%)	258 (51.7%)	103 (51.5%)	
Age (mean, SD)	69 (13.9)	68 (15.0)	69 (14.3)	0.51
Baseline NIHSS (median, IQR)	16 (11-19)	16 (11-20)	16 (10-20)	0.62
Male sex (n, %)	1291 (52.0%)	262 (52.5%)	101 (50.5%)	0.89
Pre- stroke mRS (n, %)				0.86
0	1646 (67.8%)	327 (67.7%)	134 (68.4%)	
1	324 (13.3%)	60 (12.4%)	24 (12.2%)	
2	181 (7.5%)	39 (8.1%)	11 (5.6%)	
3	155 (6.4%)	37 (7.7%)	14 (7.1%)	

4	101 (4.2%)	18 (3.7%)	10 (5.1%)	
5	22 (0.9%)	2 (0.4%)	3 (1.5%)	
Previous stroke	419 (17.0%)	77 (15.6%)	35 (17.8%)	0.69
Previous Diabetes	404 (16.4%)	78 (15.9%)	28 (14.1%)	0.68
Systolic blood pressure in mmHg (mean, SD)	150 (25.0)	148 (23.8)	152 (24.0)	0.14
Baseline ASPECTS (median, IQR)	9 (7-10)	9 (7-10)	9 (7-10)	0.21
Collateral score (n, %)				0.05
0	144 (6.3%)	34 (7.0%)	8 (4.1%)	
1	814 (35.4%)	179 (36.9%)	79 (40.9%)	
2	879 (38.2%)	203 (41.9%)	72 (37.3%)	
3	463 (20.1%)	69 (14.2%)	34 (17.6%)	
Occlusion segment (n, %)				0.52
M3 or anterior circulation	18 (0.8%)	3 (0.6%)	3 (1.5%)	
Intracranial ICA	128 (5.5%)	16 (3.3%)	11 (5.7%)	
ICA-T	501 (21.4%)	102 (20.9%)	37 (19.1%)	
M1	1357 (57.9%)	289 (59.3%)	118 (60.8%)	
M2	339 (14.5%)	77 (15.8%)	25 (12.9%)	
Thrombus imaging characteristics (median, IQR)				
Thrombus length (mm)	N/A	17 (11-29)	17 (11-26)	0.62
Thrombus perviousness (HU)	N/A	7 (14.2)	7 (15.2)	0.99
Thrombus density (HU)	N/A	49 (9.5)	50 (10.1)	0.58
IVT (n, %)	1928 (78.0%)	366 (73.6%)	133 (66.5%)	**p<0.001**
Time from on set to groin puncture in minutes (mean, SD)	205 (71.6)	192 (75.8)	200 (82.7)	**p<0.001**
First-line EVT device				0.11
Stent retriever	1354 (73.3%)	334 (72.6%)	119 (66.1%)	
Aspiration device	492 (26.7%)	126 (27.4%)	61 (33.9%)	
Post-treatment eTICI (n,%)				**p<0.001**
0	441 (18.3%)	56 (11.5%)	22 (11.2%)	
1	67 (2.8%)	17 (3.5%)	7 (3.6%)	
2A	453 (18.8%)	76 (15.6%)	44 (22.4%)	
2B	544 (22.6%)	112 (23.0%)	39 (19.9%)	
2C	235 (9.7%)	70 (14.3%)	27 (13.8%)	
3	672 (27.9%)	157 (32.2%)	57 (29.1%)	
Number of attempts to eTICI≥2B (n,%)				**p<0.01**
1	657 (37.7%)	189 (44.6%)	89 (50.9%)	
2	416 (23.9%)	101 (23.8%)	36 (20.6%)	
3	295 (16.9%)	58 (13.7%)	22 (12.6%)	

4	173 (9.9%)	41 (9.7%)	19 (10.9%)	
5+	202 (11.6%)	35 (8.3%)	9 (5.1%)	
mRS (n,%)				0.34
0	156 (6.7%)	36 (7.7%)	14 (7.4%)	
1	354 (15.3%)	73 (15.6%)	30 (16.0%)	
2	406 (17.6%)	94 (20.1%)	37 (19.7%)	
3	298 (12.9%)	72 (15.4%)	25 (13.3%)	
4	278 (12.0%)	43 (9.2%)	29 (15.4%)	
5	132 (5.7%)	21 (4.5%)	7 (3.7%)	
6	689 (29.8%)	128 (27.4%)	46 (24.5%)	

IQR: interquartile range; SD: standard deviation; NIHSS: National institute of health stroke scale; ASPECTS: Alberta stroke programme early CT score; HU: Hounsfield Units; IVT: intravenous alteplase treatment; EVT: endovascular treatment; eTICI: extended Thrombolysis in Cerebral Ischemia; mRS: modified Rankin Scale.

Supplemental Table S4. Intraclass correlation coefficient (ICC) of all thrombus radiomic features in the pilot study (n=58)

Thrombus radiomic feature	ICC	p-value	CI 95%
original_firstorder_90Percentile	0.97	<1e-9	[0.95 0.98]
original_firstorder_Energy	0.91	<1e-9	[0.86 0.95]
original_firstorder_TotalEnergy	0.91	<1e-9	[0.86 0.95]
original_gldm_DependenceNonUniformityNormalized	0.91	<1e-9	[0.84 0.94]
original_glcm_Contrast	0.90	<1e-9	[0.83 0.94]
original_glcm_DifferenceVariance	0.89	<1e-9	[0.81 0.93]
original_shape_Maximum2DDiameterColumn	0.88	<1e-9	[0.8 0.93]
original_glrlm_RunVariance	0.88	<1e-9	[0.79 0.93]
original_glcm_DifferenceAverage	0.87	<1e-9	[0.79 0.93]
original_firstorder_MeanAbsoluteDeviation	0.87	<1e-9	[0.79 0.93]
original_firstorder_RootMeanSquared	0.87	<1e-9	[0.79 0.92]
original_firstorder_InterquartileRange	0.87	<1e-9	[0.78 0.92]
original_glrlm_GrayLevelNonUniformityNormalized	0.87	<1e-9	[0.78 0.92]
original_glcm_DifferenceEntropy	0.87	<1e-9	[0.78 0.92]
original_firstorder_RobustMeanAbsoluteDeviation	0.87	<1e-9	[0.78 0.92]
original_glcm_Idm	0.87	<1e-9	[0.78 0.92]
original_glcm_Id	0.86	<1e-9	[0.77 0.92]
original_gldm_DependenceVariance	0.85	<1e-9	[0.75 0.91]
original_gldm_LargeDependenceEmphasis	0.85	<1e-9	[0.75 0.91]
original_glrlm_LongRunEmphasis	0.85	<1e-9	[0.74 0.91]
original_shape_Maximum2DDiameterSlice	0.85	<1e-9	[0.74 0.91]
original_firstorder_Entropy	0.84	<1e-9	[0.73 0.91]
original_glcm_ClusterShade	0.84	<1e-9	[0.72 0.9]
original_glcm_InverseVariance	0.83	<1e-9	[0.72 0.9]
original_glrlm_RunPercentage	0.83	<1e-9	[0.72 0.9]
original_glcm_JointEntropy	0.83	<1e-9	[0.72 0.9]
original_shape_MajorAxisLength	0.83	<1e-9	[0.71 0.9]
original_gldm_GrayLevelNonUniformity	0.83	<1e-9	[0.71 0.9]
original_shape_MeshVolume	0.82	<1e-9	[0.69 0.89]
original_shape_VoxelVolume	0.82	<1e-9	[0.69 0.89]
original_firstorder_Median	0.82	<1e-9	[0.69 0.89]
original_glrlm_ShortRunEmphasis	0.81	<1e-5	[0.68 0.89]

original_firstorder_Mean	0.81	<1e-5	[0.68 0.89]
original_glrlm_RunEntropy	0.81	<1e-5	[0.67 0.89]
original_glcm_SumEntropy	0.81	<1e-5	[0.67 0.88]
original_glrlm_RunLengthNonUniformityNormalized	0.80	<1e-5	[0.67 0.88]
original_shape_SurfaceArea	0.80	<1e-5	[0.66 0.88]
original_glrlm_GrayLevelNonUniformity	0.80	<1e-5	[0.66 0.88]
original_glcm_SumSquares	0.80	<1e-5	[0.66 0.88]
original_firstorder_Uniformity	0.79	<1e-5	[0.65 0.88]
original_glrlm_RunLengthNonUniformity	0.79	<1e-5	[0.65 0.88]
original_glcm_Correlation	0.78	<1e-5	[0.63 0.87]
original_glszm_ZonePercentage	0.78	<1e-5	[0.62 0.87]
original_glszm_LargeAreaLowGrayLevelEmphasis	0.78	<1e-5	[0.62 0.87]
original_gldm_DependenceEntropy	0.77	<1e-5	[0.62 0.87]
original_gldm_DependenceNonUniformity	0.77	<1e-5	[0.62 0.87]
original_glcm_Idmn	0.77	<1e-5	[0.61 0.86]
original_glcm_ClusterTendency	0.76	<1e-5	[0.6 0.86]
original_glcm_Imc2	0.76	<1e-5	[0.59 0.86]
original_shape_LeastAxisLength	0.76	<1e-5	[0.59 0.86]
original_glszm_ZoneEntropy	0.76	<1e-5	[0.59 0.86]
original_glcm_JointEnergy	0.75	<1e-5	[0.58 0.85]
original_shape_SurfaceVolumeRatio	0.74	<1e-5	[0.57 0.85]
original_gldm_SmallDependenceEmphasis	0.74	<1e-5	[0.56 0.84]
original_glszm_SizeZoneNonUniformity	0.74	<1e-5	[0.55 0.84]
original_glcm_Imc1	0.73	<1e-5	[0.55 0.84]
original_shape_Maximum2DDiameterRow	0.73	<1e-5	[0.55 0.84]
original_glszm_ZoneVariance	0.73	<1e-5	[0.54 0.84]
original_ngtdm_Busyness	0.72	<1e-5	[0.53 0.83]
original_gldm_LargeDependenceLowGrayLevelEmphasis	0.72	<1e-5	[0.52 0.83]
original_glcm_MaximumProbability	0.72	<1e-5	[0.52 0.83]
original_glcm_MCC	0.71	<1e-5	[0.51 0.83]
original_shape_MinorAxisLength	0.71	<1e-5	[0.51 0.83]
original_shape_Flatness	0.71	<1e-5	[0.51 0.83]
original_glrlm_LongRunLowGrayLevelEmphasis	0.71	<1e-5	[0.51 0.83]
original_glszm_LargeAreaEmphasis	0.70	<1e-5	[0.5 0.82]
original_glcm_ClusterProminence	0.68	<1e-3	[0.45 0.81]

original_glcm_Idn	0.67	<1e-3	[0.45 0.81]
original_ngtdm_Contrast	0.67	<1e-3	[0.43 0.8]
original_shape_Elongation	0.66	<1e-3	[0.42 0.8]
original_firstorder_Variance	0.62	<1e-3	[0.36 0.78]
original_gldm_GrayLevelVariance	0.58	<1e-3	[0.28 0.75]
original_firstorder_Maximum	0.57	<1e-3	[0.28 0.75]
original_firstorder_10Percentile	0.56	<0.05	[0.26 0.74]
original_gldm_LowGrayLevelEmphasis	0.55	<0.05	[0.25 0.74]
original_glszm_GrayLevelNonUniformity	0.55	<0.05	[0.24 0.73]
original_shape_Maximum3DDiameter	0.52	<0.05	[0.19 0.72]
original_ngtdm_Coarseness	0.52	<0.05	[0.19 0.72]
original_glrlm_LowGrayLevelRunEmphasis	0.50	<0.05	[0.16 0.71]
original_shape_Sphericity	0.50	<0.05	[0.15 0.7]
original_glszm_LowGrayLevelZoneEmphasis	0.42	<0.05	[0.01 0.65]
original_glrlm_GrayLevelVariance	0.39	<0.05	[-0.02 0.64]
original_glszm_GrayLevelNonUniformityNormalized	0.39	<0.05	[-0.03 0.64]
original_glrlm_ShortRunLowGrayLevelEmphasis	0.39	<0.05	[-0.04 0.64]
original_gldm_SmallDependenceLowGrayLevelEmphasis	0.36	<0.05	[-0.09 0.62]
original_glszm_SmallAreaEmphasis	0.26	0.13	[-0.24 0.56]
original_glszm_SizeZoneNonUniformityNormalized	0.20	0.20	[-0.34 0.53]
original_ngtdm_Complexity	0.15	0.27	[-0.43 0.5]
original_glszm_SmallAreaLowGrayLevelEmphasis	0.09	0.36	[-0.54 0.46]
original_firstorder_Range	0.07	0.40	[-0.58 0.45]
original_firstorder_Skewness	0.01	0.48	[-0.67 0.42]
original_glcm_JointAverage	0.01	0.48	[-0.67 0.41]
original_glcm_SumAverage	0.01	0.48	[-0.67 0.41]
original_gldm_SmallDependenceHighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]
original_glrlm_LongRunHighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]
original_glszm_SmallAreaHighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]
original_glrlm_ShortRunHighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]
original_gldm_LargeDependenceHighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]
original_glrlm_HighGrayLevelRunEmphasis	0.00	0.50	[-0.69 0.41]
original_glcm_Autocorrelation	0.00	0.50	[-0.69 0.41]
original_firstorder_Kurtosis	0.00	0.50	[-0.69 0.41]
original_gldm_HighGrayLevelEmphasis	0.00	0.50	[-0.69 0.41]

original_glszm_LargeAreaHighGrayLevelEmphasis	0.00	0.51	[-0.7 0.41]
original_glszm_GrayLevelVariance	0.00	0.51	[-0.7 0.41]
original_glszm_HighGrayLevelZoneEmphasis	0.00	0.51	[-0.7 0.41]
original_ngtdm_Strength	-0.04	0.56	[-0.76 0.38]
original_firstorder_Minimum	-0.04	0.56	[-0.76 0.38]

A description of all features is available here:
https://pyradiomics.readthedocs.io/en/latest/features.html

Supplemental Table S5. **Training set performance**

	eTICI≥2B	First-attempt eTICI≥2B	eTICI≥2B within three attempts	mRS≤2
CV+MTM+TR	0.71(0.06)	0.64(0.06)	**0.74(0.04)**	**0.8(0.04)**
CV+TR	**0.68(0.05)**	**0.69(0.06)**	0.65(0.05)	0.78(0.04)
CV+MTM	0.67(0.05)	0.62(0.06)	0.62(0.05)	0.79(0.05)
CV	0.63(0.05)	0.64(0.06)	0.64(0.06)	0.79(0.05)
MTM	0.61(0.05)	0.6(0.06)	0.58(0.06)	0.62(0.05)
TR	0.61(0.06)	0.6(0.06)	0.59(0.06)	0.57(0.06)

Results of the area under the receiver operating characteristic curve per variable selection set. Values are presented as mean (standard deviation) over 100 cohorts of 100 patients resampled with replacement from the training set. Clinical variables (CV), manual thrombus measures (MTM), thrombus radiomics (TR), modified Rankin Scale (mRS), the extended Thrombolysis in Cerebral Ischemia (eTICI) score.

Supplemental Table S6. **Extensive evaluation metrics for test set results**

Outcome	Variables	AUC	TPR	TNR	PPV	NPV	Brier
mRS≤2	CV+MTM+TR	0.79 (0.04)	0.29 (0.06)	0.92 (0.04)	0.74 (0.11)	0.63 (0.05)	0.22 (0.01)
mRS≤2	CV+TR	0.78 (0.04)	0.13 (0.05)	0.97 (0.02)	0.78 (0.15)	0.6 (0.05)	0.22 (0.01)
mRS≤2	CV+MTM	0.81 (0.05)	0.3 (0.06)	0.93 (0.03)	0.77 (0.11)	0.63 (0.05)	0.21 (0.01)
mRS≤2	CV	0.81 (0.04)	0.81 (0.06)	0.74 (0.05)	0.69 (0.06)	0.85 (0.05)	0.21 (0.01)
mRS≤2	MTM	0.57 (0.05)	0.0 (0.0)	1.0 (0.0)	0 (0)	0.57 (0.05)	0.24 (0.01)
mRS≤2	TR	0.50 (0.07)	0.0 (0.0)	1.0 (0.0)	0 (0)	0.57 (0.06)	0.25 (0.01)
eTICI≥2B	CV+MTM+TR	0.54 (0.06)	1.0 (0.0)	0.0 (0.0)	0.63 (0.05)	0 (0)	0.23 (0.02)
eTICI≥2B	CV+TR	0.52 (0.05)	1.0 (0.0)	0.0 (0.0)	0.64 (0.05)	0 (0)	0.23 (0.02)
eTICI≥2B	CV+MTM	0.53 (0.06)	0.54 (0.07)	0.49 (0.09)	0.65 (0.06)	0.38 (0.08)	0.25 (0.0)
eTICI≥2B	CV	0.56 (0.06)	1.0 (0.0)	0.0 (0.0)	0.64 (0.05)	0 (0)	0.23 (0.02)
eTICI≥2B	MTM	0.5 (0.06)	0.54 (0.06)	0.45 (0.09)	0.64 (0.07)	0.35 (0.07)	0.25 (0.0)
eTICI≥2B	TR	0.52 (0.06)	1.0 (0.0)	0.0 (0.0)	0.63 (0.05)	0 (0)	0.24 (0.02)
First-pass eTICI≥2B	CV+MTM+TR	0.56 (0.06)	0.57 (0.08)	0.55 (0.07)	0.44 (0.08)	0.66 (0.07)	0.25 (0.0)
First-pass eTICI≥2B	CV+TR	0.58 (0.07)	0.0 (0.0)	1.0 (0.0)	0 (0)	0.61 (0.05)	0.24 (0.01)
First-pass eTICI≥2B	CV+MTM	0.51 (0.06)	0.41 (0.08)	0.59 (0.06)	0.39 (0.08)	0.61 (0.06)	0.25 (0.0)
First-pass eTICI≥2B	CV	0.50 (0.06)	0.43 (0.08)	0.49 (0.07)	0.34 (0.08)	0.58 (0.07)	0.25 (0.0)
First-pass eTICI≥2B	MTM	0.53 (0.06)	0.40 (0.08)	0.61 (0.05)	0.39 (0.08)	0.63 (0.06)	0.25 (0.0)
First-pass eTICI≥2B	TR	0.59 (0.07)	0.0 (0.0)	1.0 (0.0)	0 (0)	0.62 (0.05)	0.24 (0.01)

eTICI≥2B within three passes	CV+MTM+TR	0.60 (0.05)	1.0 (0.0)	0.0 (0.0)	0.55 (0.05)	0 (0)	0.24 (0.01)
eTICI≥2B within three passes	CV+TR	0.58 (0.06)	0.59 (0.07)	0.48 (0.07)	0.59 (0.06)	0.49 (0.07)	0.25 (0.0)
eTICI≥2B within three passes	CV+MTM	0.57 (0.05)	0.52 (0.07)	0.52 (0.07)	0.57 (0.07)	0.46 (0.07)	0.25 (0.0)
eTICI≥2B within three passes	CV	0.55 (0.07)	0.47 (0.07)	0.55 (0.08)	0.57 (0.08)	0.45 (0.07)	0.25 (0.0)
eTICI≥2B within three passes	MTM	0.53 (0.05)	0.51 (0.07)	0.57 (0.07)	0.59 (0.07)	0.49 (0.07)	0.25 (0.0)
eTICI≥2B within three passes	TR	0.61 (0.06)	1.0 (0.0)	0.0 (0.0)	0.55 (0.05)	0 (0)	0.25 (0.01)

AUC: area under the receiver operating characteristic curve, TPR: true positive rate (sensitivity), TNR: True negative rate (specificity), PPV: positive predictive value (precision), NPV: negative predictive value. CV: clinical variables, MTM: manual thrombus measurements, TR: Thrombus radiomics; Brier: brier score, the average difference between the predicted probability and the ground truth class. Values are presented as mean (standard deviation) over 100 cohorts of 100 patients resampled with replacement from the test set.

Supplemental References

1. Jansen IGH, Mulder M, Goldhoorn RB, investigators MCR. Endovascular treatment for acute ischaemic stroke in routine clinical practice: prospective, observational cohort study (MR CLEAN Registry). *BMJ*. 2018;360:k949. doi: 10.1136/bmj.k949
2. Barber PA, Demchuk AM, Zhang J, Buchan AM. Validity and reliability of a quantitative computed tomography score in predicting outcome of hyperacute stroke before thrombolytic therapy. ASPECTS Study Group. Alberta Stroke Programme Early CT Score. *Lancet*. 2000;355:1670-1674. doi: 10.1016/s0140-6736(00)02237-6
3. Tan IY, Demchuk AM, Hopyan J, Zhang L, Gladstone D, Wong K, Martin M, Symons SP, Fox AJ, Aviv RI. CT angiography clot burden score and collateral score: correlation with clinical and radiologic outcomes in acute middle cerebral artery infarct. *AJNR Am J Neuroradiol*. 2009;30:525-531. doi: 10.3174/ajnr.A1408

4. Tung EL, McTaggart RA, Baird GL, Yaghi S, Hemendinger M, Dibiasio EL, Hidlay DT, Tung GA, Jayaraman MV. Rethinking Thrombolysis in Cerebral Infarction 2b: Which Thrombolysis in Cerebral Infarction Scales Best Define Near Complete Recanalization in the Modern Thrombectomy Era? *Stroke*. 2017;48:2488-2493. doi: 10.1161/STROKEAHA.117.017182